

Part 4: NLP Interview Cheatsheet (8 Essential Must-Know Foundations)

A high-density revision reference for AI, Applied AI, GenAI, and ML Engineer interviews covering the 8 essential formulas, core diagrams, time & memory complexity, interview Q&A, and a 1-page comparison table.

1. The 8 Must-Know Essential Formulas

1. TF-IDF Formula:

$$TF(t, d) = \text{count}(t, d) / |d|$$
$$IDF(t) = \log((1 + |D|) / (1 + DF(t))) + 1$$
$$TF-IDF(t, d) = TF(t, d) * IDF(t)$$

2. Cosine Similarity:

$$\text{CosineSim}(u, v) = (u \cdot v) / (||u||_2 * ||v||_2)$$

3. Naive Bayes (+ Laplace Add-1 Smoothing):

$$P(c \mid d) \propto P(c) * \prod P(w_i \mid c)$$
$$P(w_i \mid c) = (N_{\{c,i\}} + 1) / (N_c + |V|)$$

4. Word2Vec Skip-Gram Sigmoid Output:

$$P(w_O \mid w_I) = \sigma(v'_{w_O}^T v_{w_I}) = 1 / (1 + \exp(-v'_{w_O}^T v_{w_I}))$$

5. LSTM Cell Update Equations (Forward Pass Only):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (\text{Forget Gate})$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (\text{Input Gate})$$
$$C_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (\text{Candidate Cell})$$
$$C_t = f_t \odot C_{t-1} + i_t \odot C_t \quad (\text{Cell State Update})$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (\text{Output Gate})$$
$$h_t = o_t \odot \tanh(C_t) \quad (\text{Hidden State})$$

6. Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}((Q K^T) / \sqrt{d_k}) V$$

7. BLEU Score (BLEU-2 Precision-Oriented):

$$\text{BLEU-2} = BP * \exp(0.5 \log p_1 + 0.5 \log p_2)$$
$$BP = \exp(1 - r/c) \text{ if } c \leq r \text{ else } 1.0$$

8. Perplexity (Model Uncertainty Branch Factor):

$$\text{PPL} = \exp(L_{\{\text{CrossEntropy}\}})$$

2. Time & Memory Complexity Matrix

Algorithm / Architecture	Training Time Complexity	Inference Time Complexity	Memory Footprint Complexity

TF-IDF Indexing	$O(D \times L)$	$O(k \times V)$	$O(D \times V)$ Sparse
Multinomial Naive Bayes	$O(D \times L)$	$O(C \times L)$	$O(C \times V)$ Dense
Word2Vec Skip-Gram	$O(C \times m \times K \times d)$	$O(1)$ Lookup	$O(V \times d)$ Embedding Table
LSTM Model Layer	$O(T \times d^2)$ per sequence	$O(T \times d^2)$	$O(4 \times d^2)$ Weights
Scaled Dot-Product Attention	$O(T^2 \times d)$ per sequence	$O(T^2 \times d)$	$O(T^2)$ RAM Attention Weights
INT8 Quantized Model	$O(N \times d)$ SIMD	$O(N \times d)$ SIMD	$O(\frac{1}{4} \times M_{\text{FP32}})$ Bytes

3. High-Frequency Interview Q&A Summary

1. Why do we scale $QK^\top$ by $\sqrt{d_k}$ in Attention?

- *Answer:* Large d_k causes dot products $q \cdot k$ to have variance d_k . Large dot products push Softmax into saturated regions with near-zero gradients ($\text{softmax}'(z) \rightarrow 0$), causing vanishing gradients during backpropagation. Dividing by $\sqrt{d_k}$ rescales variance back to 1.0.

2. Why does Laplace Smoothing prevent zero probability in Naive Bayes?

- *Answer:* Unseen vocabulary words in a training class yield raw likelihood $P(w_{\text{new}} | c) = 0$, causing the entire posterior product to collapse to zero. Adding $+1$ ensures every word retains a positive non-zero likelihood $P(w_i | c) = \frac{N_{c,i}+1}{N_c+|V|}$.

3. How does the LSTM cell state C_t resolve vanishing gradients?

- *Answer:* Cell update $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$ is additive. When forget gate $f_t = 1$, the gradient derivative $\frac{\partial C_t}{\partial C_{t-1}} = 1$, preserving gradient norm ≈ 1.0 across long sequences.

4. Why is lower Perplexity better?

- *Answer:* $\text{PPL} = \exp(\mathcal{L}_{\text{CE}})$. Perplexity represents the average branching factor of uncertainty when predicting the next token. A lower PPL means fewer candidate choices the model is uncertain between.

5. Why is BLEU precision-oriented while ROUGE is recall-oriented?

- *Answer:* BLEU checks what fraction of generated candidate n-grams are valid (preventing hallucinated ungrounded text in translation). ROUGE checks what fraction of reference target tokens were captured in the summary (preventing key detail omission).